



## Special Topic in Computer Science (CSC649)

### Big Data and Machine Learning Project

#### Premium VS Freemium Streaming Platform Preferences

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## **ABSTRACT**

The growing preference for freemium and premium streaming platforms among users necessitates a detailed exploration of preferences and satisfaction levels related to these services. In an era increasingly shaped by digital solutions, understanding the unique needs of users is crucial for these platforms to tailor their offerings effectively. This research delves into the multifaceted factors influencing the selection of freemium versus premium streaming services, encompassing considerations such as convenience, content variety, and overall user experience. The study aims to provide invaluable guidance to streaming service providers, facilitating the optimization of their services to align with the discerning expectations of their user base. Addressing potential obstacles faced by these platforms, including advertising intrusiveness in freemium models and content accessibility in premium models, the research aspires to contribute to a more tailored and satisfactory streaming experience for users. The implementation of a K-Nearest Neighbors (KNN) classification model further validates the predictive capabilities of the study, showcasing high testing accuracy and robust performance across diverse subsets of the dataset. The findings not only offer practical insights for industry players but also contribute to academic knowledge in consumer behavior and technology adoption within specific demographics. The research holds significance beyond the immediate context, serving as a potential model for similar studies in other sectors and guiding the enhancement of user experiences in the dynamic landscape of streaming services.

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## LIST OF ABBREVIATIONS

|     |                        |
|-----|------------------------|
| CSV | Comma-Separated Values |
| KNN | K-Nearest Neighbour    |

## **1.0 Introduction**

The growing reliance on streaming platforms in student's daily lives at UiTM Jasin underscores the need for a thorough investigation of the specific preferences and satisfaction levels linked with two types of services: premium and freemium streaming platforms. In an era where digital solutions influence customer decisions, understanding this population's specific demands is critical for these platforms to properly personalize their offerings. The purpose of this study is to investigate the many factors influencing UiTM Jasin students' choice of streaming services, considering aspects such as convenience, content diversity, and overall user experience. By collecting comprehensive insights into the complex dimensions of their preferences, this study hopes to provide helpful recommendations to both premium and freemium streaming services, assisting in the improvement of their offers to better satisfy the sophisticated expectations of UiTM Jasin students. Given the current landscape's challenges and opportunities, the research aims not only to uncover detailed preferences, but also to address potential barriers to these platforms meeting the unique demands of the UiTM Jasin community, such as subscription costs, content accessibility, and platform usability. In doing so, the study hopes to contribute to a more tailored and enjoyable streaming experience for this demographic.

## **2.0 Problem Statement**

As people become more reliant on streaming platforms, there is a need to look deeper into their specific preferences and satisfaction levels with two types of services, premium and freemium streaming platforms. Understanding the specific requirements of this demographic is critical for these platforms to effectively cater to their preferences. The purpose of this study is to investigate the factors that influence individuals' choice of streaming platforms, including convenience, content variety, and overall user experience. By learning about the subtleties of their tastes, the study hopes to provide useful information that will help both premium and freemium streaming services optimize their products to better meet the expectations of customers.

The current landscape of streaming platforms poses challenges and opportunities in meeting the distinct requirements of individuals. One significant problem lies in the lack of tailored services that specifically address the preferences of this demographic, potentially leading to dissatisfaction and suboptimal user experiences. Additionally, issues such as subscription costs, content accessibility, and platform usability might present obstacles that hinder the seamless integration of streaming services into the daily lives of users. Therefore, this research not only aims to uncover the nuanced preferences of this population but also seeks to identify and address potential challenges faced by premium and freemium streaming services in meeting the unique demands of the user.

### **3.0 Project Objective**

This research project aims to:

1. To identify UiTM Jasin students preferred streaming platforms, including premium and freemium services.
2. Determine the level of satisfaction among UiTM Jasin students with premium and freemium streaming platforms.
3. Investigate UiTM Jasin users' specific preferences for convenience, content variety, and overall user experience when using streaming services.

### **4.0 Project Scope**

The goal of this research project is to get a full overview of the streaming platform environment among UiTM Jasin students. The scope includes conducting surveys and interviews with university students to acquire information on their streaming platform preferences, satisfaction levels, and issues, with an emphasis on premium and freemium services. The investigation will look at crucial criteria including convenience, content diversity, user experience, subscription costs, and platform usability. The study's scope is limited to the unique setting of UiTM Jasin, ensuring that the insights gained are distinct and relevant to the requirements of this community. By focusing on these elements, the study hopes to provide a more detailed and targeted understanding of UiTM Jasin individuals' streaming service preferences and experiences.

### **5.0 Significance of Project**

This study has significant implications for both academic knowledge and practical implementations in the streaming platform sector. The study adds to the academic knowledge base on consumer behavior and technology adoption within specific demographics by investigating the preferences and obstacles faced by UiTM Jasin persons when using premium and freemium streaming services. Furthermore, the findings provide useful insights for industry players, allowing them to optimize their services to better meet the expectations of UiTM Jasin users. The project's importance goes beyond its immediate context, as it can serve as a model for comparable studies in other educational institutions, assisting in improving user experiences and refining services for diverse populations across multiple geographies.

### **6.0 Literature Review**

The debate over user preferences between freemium and premium streaming services is critical for understanding consumer behavior in digital content services. Freemium models offer basic services free of charge while charging for enhanced features, aiming to convert free users into paying subscribers. Emotional value, such as the enjoyment derived from using the service, is a significant predictor of the intention to upgrade. Users who find enjoyment in the freemium service often desire the enhanced experience offered by the premium service. Functional value, particularly



the ability to access the service anytime and anywhere (ubiquity), significantly affects the retention of premium subscriptions. Users highly value the convenience and seamless access provided by premium services.

Additionally, the perceived price value of the premium subscription influences both the intention to upgrade and the retention of the subscription, as users are likely to upgrade if they perceive the premium service as offering good value for the price. Interestingly, social connectivity has been found to have no significant effect on the intention to upgrade but a small negative effect on the intention to retain the premium subscription. This suggests that while social features may not drive initial upgrades, they might influence long-term satisfaction and retention negatively if users do not find enough social value in the premium service. The intrusiveness of advertisements in the freemium version can negatively affect the perceived value of the premium subscription. If free users find ads too intrusive, it could diminish their overall experience, potentially reducing their willingness to upgrade to premium. Moreover, the ability to discover new content is a strong driver for retaining premium subscriptions. Users value premium services that offer exclusive or new content that is not available in the free version. This continuous content discovery can enhance user satisfaction and encourage long-term retention. The decision to upgrade from freemium to premium is primarily driven by the enjoyment and perceived value of the premium service, with immediate benefits such as an ad-free experience and enhanced features motivating free users to pay for the premium version. On the other hand, retention of premium subscriptions is driven by factors that enhance long-term user satisfaction, such as the convenience of ubiquitous access and the continuous discovery of new content, which contribute to sustained user engagement and justify the recurring cost of the premium service. Freemium and premium users have different standards and expectations. While freemium users might tolerate certain limitations and intrusiveness due to the no-cost nature of the service, premium users expect a seamless, high-quality experience that justifies their payment.

The decision to upgrade represents a departure from the user's status quo and is influenced by immediate gratifications, whereas the decision to retain is based on sustained value perception over time. Understanding the distinct factors influencing the transition from freemium to premium and the retention of premium subscriptions can help digital content service providers tailor their offerings and marketing strategies. Emotional, functional, and economic values play crucial roles in user decisions, while the continuous delivery of new content and a seamless experience are essential for retaining premium users. Recognizing these dynamics is vital for maximizing revenue and maintaining a loyal user base in the competitive streaming market. This literature review is based on insights derived from the study by Mäntymäki et al. (2020), which provides a comprehensive analysis of consumer behavior in the context of freemium digital content service.

## **7.0 Data Collection**

The dataset was carefully imported using a designated CSV file directory. It is the result of a thorough survey that was carried out with 500 respondents using Google Forms. As can be seen in the image below, a thorough analysis of the dataset yields six numerical and four categorical. Figure 1.1 presents the collection of datasets.

|    |                                                                                                                                  |
|----|----------------------------------------------------------------------------------------------------------------------------------|
| 1  | Gender, Age, MoneySpent, PlatformSubscribed, TimeSpent, PremiumRating, FreemiumRate, DeviceType, ContentType, PreferredPlatform  |
| 2  | Female, 47, 100, 5, 2 hours, 4, 4, Smart TV, Documentaries, "Premium (Netflix, Disney Hotstar, Prime, etc)"                      |
| 3  | Male, 54, 60.00, 2, 3 hours per week, 3, 3, Smart TV, Sport Programming, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)" |
| 4  | Male, 25, 100, 100, 10 hours, 5, 5, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                               |
| 5  | Male, 37, 300, 3, 4 hours, 5, 5, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                  |
| 6  | Female, 20, 100, 1, 12 hours, 5, 5, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                               |
| 7  | Female, 23, 98, 3, 6 hours, 5, 2, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                |
| 8  | Male, 25, 50, 50, 10 hours, 5, 5, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                |
| 9  | Female, 19, 100, 5, 5 hours a day, 3, 3, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                         |
| 10 | Female, 19, 0, 5, 1 hours, 3, 3, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                  |
| 11 | Female, 22, 10, 1, 3, 5, 3, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                      |
| 12 | Female, 19, 10, 10, 5 hours, 4, 2, Desktop Computer, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                   |
| 13 | Male, 26, 17, 1, 2 hours per day, 4, 4, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                        |
| 14 | Female, 25, 30, 2, 1 hour, 4, 4, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                  |
| 15 | Male, 24, 17, 2, 3 hour, 4, 4, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                    |
| 16 | Female, 49, 60, 2, 1 hour, 4, 4, Phone, Documentaries, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                   |
| 17 | Male, 35, 500, 2, 4, 4, 4, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                        |
| 18 | Female, 51, 180, 2, 3, 4, 3, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                   |
| 19 | Male, 43, 0, 0, 0, 3, 3, Smart TV, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                               |
| 20 | Female, 49, 50, 2, 3, 4, 4, Phone, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                     |
| 21 | Male, 18, 30, 1, 1 hour, 4, 4, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                    |
| 22 | Female, 24, 70, 2, 2 hr, 4, 3, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                    |
| 23 | Male, 17, 35, 3, 0, 5, 5, Smart TV, Sport Programming, "Premium (Netflix, Disney Hotstar, Prime, etc)"                           |
| 24 | Male, 23, 150, 3, 5hrs, 5, 5, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                     |
| 25 | Female, 28, 30, 1, 2 hrs per day, 3, 4, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                        |
| 26 | Female, 55, 30.00, 1, 4 hours, 3, 4, Phone, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                      |
| 27 | Male, 18, 0, 0, 1, 3, 4, Desktop Computer, TV shows, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                     |
| 28 | Female, 52, 70, 3, 2 jam, 4, 4, Desktop Computer, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                      |
| 29 | Female, 51, 200, 3, 2 hours, 4, 4, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                             |
| 30 | Female, 26, 70, 2, 2 hours, 3, 3, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                |
| 31 | Female, 26, 10, 1, 3 times, 4, 4, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                |
| 32 | Female, 20, 0, 0, 1 hour a day, 4, 5, Desktop Computer, TV shows, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"        |
| 33 | Female, 24, 20, 2, 4 hours/day, 5, 3, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                            |
| 34 | Female, 22, 0, 0, 3, 3, 5, Phone, Music, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                                 |
| 35 | Female, 22, 5, 1, 20, 5, 5, Tablet, Sport Programming, "Premium (Netflix, Disney Hotstar, Prime, etc)"                           |
| 36 | Female, 22, 49, 2, 5, 5, 3, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                    |
| 37 | Male, 21, 0, 0, 2, 5, 4, Phone, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                                  |
| 38 | Female, 25, 25, 5, 2 or 3 hours, 5, 4, Smart TV, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                         |
| 39 | Male, 22, 5, 1, 15, 4, 4, Phone, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                                 |
| 40 | Female, 23, 11, 1, 6, 5, 3, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                      |
| 41 | Female, 22, 15, 2, 12, 5, 5, Phone, Music, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                       |
| 42 | Male, 22, 0, 0, 2-3 hours, 5, 3, Desktop Computer, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"               |
| 43 | Female, 22, 0, 1, 0, 3, 4, Desktop Computer, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                     |
| 44 | Female, 22, 20, 3, 2 hours, 4, 5, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                |
| 45 | Male, 22, 4, 1, 2 hours, 3, 2, Tablet, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                   |
| 46 | Female, 22, 0, 0, 2 hours, 5, 5, Desktop Computer, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"               |
| 47 | Female, 22, 0, 5, 4-5hr, 5, 4, Phone, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                    |
| 48 | Male, 27, 399.90, 1, 24 hours, 4, 3, Desktop Computer, Movies, "Premium (Netflix, Disney Hotstar, Prime, etc)"                   |
| 49 | Male, 22, 4, 1, 2 hour per day, 5, 2, Phone, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                           |
| 50 | Male, 13, 50, 45, 3, 5, 5, Phone, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                      |
| 51 | Male, 13, 50, 45, 3, 5, 5, Phone, TV shows, "Premium (Netflix, Disney Hotstar, Prime, etc)"                                      |
| 52 | Male, 22, 0, 0, 1, 1, 5, Phone, Movies, "Freemium (123 Movies, Standard Youtube, Fmovies, etc)"                                  |

Figure 1.1 Dataset Collection

## 7.1 Data Analytic Purposes

Our survey consists of 10 questions, including 4 categorical and 6 numerical questions. The first question asks, "What is your gender?" followed by, "Please state your age." Next, participants are asked, "How much money (in RM) do you spend on premium streaming per month?" and "How much time (in hours) do you spend on streaming platforms?" Additionally, there are questions asking participants to rate their experience with premium streaming platforms and freemium streaming platforms on a linear scale from 1 (Very Bad) to 5 (Very Good). Further questions inquire about device preferences for streaming content, with options including Phone, Desktop Computer, Tablet, Smart TV, and Other. Participants are also asked, "What type of content do you prefer?" with multiple choice options such as Movies, TV Shows, Documentaries, Sports Programming, and Other. The final question, which is our key focus, asks participants to indicate their preference between freemium and premium streaming platforms. All the survey details are illustrated in the figure below.

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>df.info()</code> | <pre> &lt;class 'pandas.core.frame.DataFrame'&gt; RangeIndex: 500 entries, 0 to 499 Data columns (total 10 columns): #   Column                Non-Null Count  Dtype ---  - 0   Gender                500 non-null   object 1   Age                  500 non-null   int64 2   MoneySpent           500 non-null   float64 3   PlatformSubscribed   500 non-null   int64 4   TimeSpent            500 non-null   object 5   PremiumRating        500 non-null   int64 6   FreemiumRate         500 non-null   int64 7   DeviceType           500 non-null   object 8   ContentType          500 non-null   object 9   PreferredPlatform    500 non-null   object dtypes: float64(1), int64(4), object(5) memory usage: 39.2+ KB </pre> |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 1.2 Data information

### 7.1.1 Categorical Data

The tabulation of categorical data, as illustrated in Figure 1.3, includes 5 variables, all characterized by their 'object' data types. In the context of pandas DataFrames, the 'object' data type commonly indicates categorical data, implying that these variables likely denote non-numeric information or categorical attributes within the dataset.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> &lt;class 'pandas.core.frame.DataFrame'&gt; RangeIndex: 400 entries, 0 to 399 Data columns (total 5 columns): #   Column                Non-Null Count  Dtype ---  - 0   Gender                400 non-null   object 1   TimeSpent            400 non-null   object 2   DeviceType           400 non-null   object 3   ContentType          400 non-null   object 4   PreferredPlatform    400 non-null   object dtypes: object(5) memory usage: 15.8+ KB None </pre> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 1.3 List of Columns with Object Data Type

### 7.1.2 Numerical Data

Figure below shows the numerical data, with all of them designated as 'int64' or data types. These variables encapsulate quantitative details or measurements within the dataset, suggesting the likelihood of numeric attributes involving integers.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   400 non-null   int64
1   PlatformSubscribed    400 non-null   int64
2   PremiumRating         400 non-null   int64
3   FreemiumRate          400 non-null   int64
dtypes: int64(4)
memory usage: 12.6 KB
None

```

Figure 1.4 List of Columns with Integer Data Type

## 7.2 Data Science Purposes

In this sense, data science refers to emphasizing objectives or goals associated with data analysis. The two key components that make this possible are "Label" and "Features." Every component is essential to comprehending and using data in an efficient manner. "Features" are the characteristics of the data that are used to make predictions, whereas "Label" is the expected category or value of the data.

### 7.2.1 Label

Label is the predicted category which means it is the target variable, the outcome we are aiming to predict. It is the specific feature in a dataset that holds the key information we want to understand or forecast. In this project, the label is the preferred platform, which consists of Premium or Freemium.

### 7.2.2 Features

The independent variables in a dataset that provide insight into the target variable, "label," are called features. Another name for it is a non-target column. By analysing the patterns and connections between these features and the label, the model gains the ability to make precise predictions and classifications.

## 7.3 Target Respondent

Students who were selecting between Premium and Freemium streaming platforms were the project's target audience. 500 respondents participated in a data gathering phase to guarantee reliable and meaningful results. This extensive sample size enables thorough analysis and a more profound comprehension of user preferences within these platforms.

### 7.3.1 Data Collection Instrument

This dataset was collected through a structured questionnaire via Google Form.

### 7.3.2 Distribution Strategy

To reach a broad range of participants within a limited timeframe, the questionnaire was spread across various platforms like Instagram, WhatsApp, Facebook and Telegram. Data collection spanned from May 6th to June 5th, 2024.

## 8.0 Data Analysis

In this section, we delve into a comprehensive analysis of the data to extract meaningful insights and patterns. The analysis is divided into two primary parts: Categorical Data Analysis and Numerical Data Analysis. Each part involves the creation and discussion of a variety of charts to visualize the data effectively.

### 8.1 Categorical Data Analysis

Categorical data analysis focuses on the non-numeric attributes of the dataset, such as gender, device type, content preferences, and platform choices. Through various types of charts, we can visualize the distribution and frequency of these categorical variables, providing insights into user demographics and preferences.

#### 1. Gender Distribution

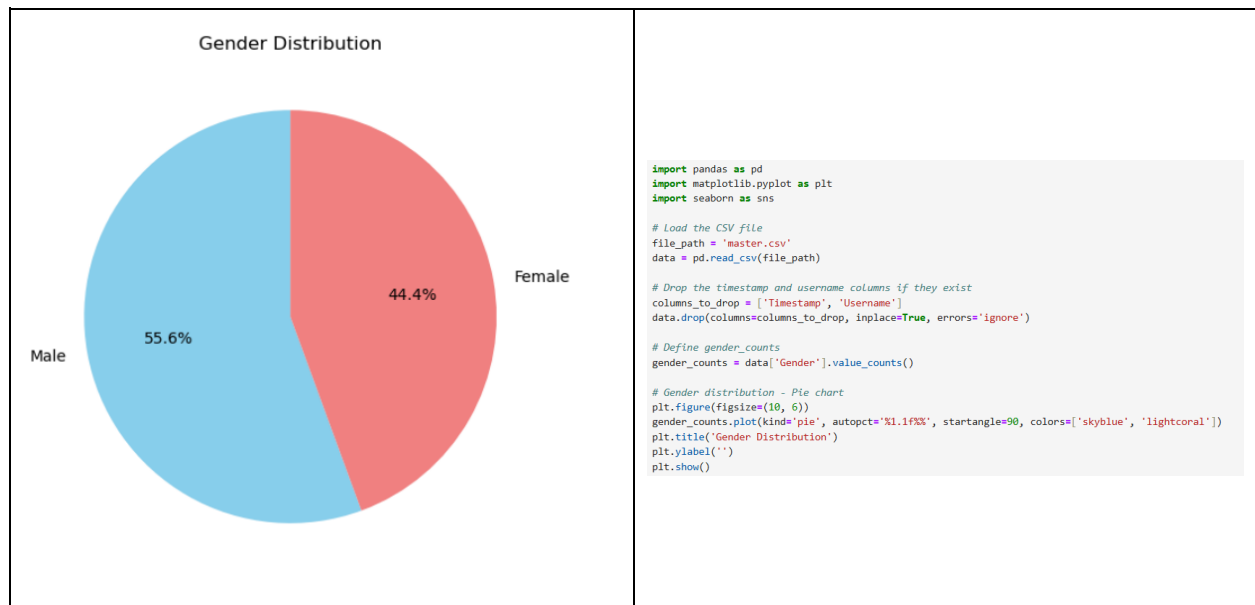


Figure 2.1 Pie Chart of Gender Distribution

This pie chart represents the gender distribution of respondents. By visualizing the data in a pie chart, we can quickly see the proportion of male and female participants in the dataset. Each segment of the pie chart is proportional to the number of respondents of each gender. The percentages indicate the share of each gender, making it easy to understand the gender balance. This visualization is helpful for identifying any gender biases or disparities in the sample population, which can be crucial for studies aiming for gender diversity and representation.

2. Device Type Distribution

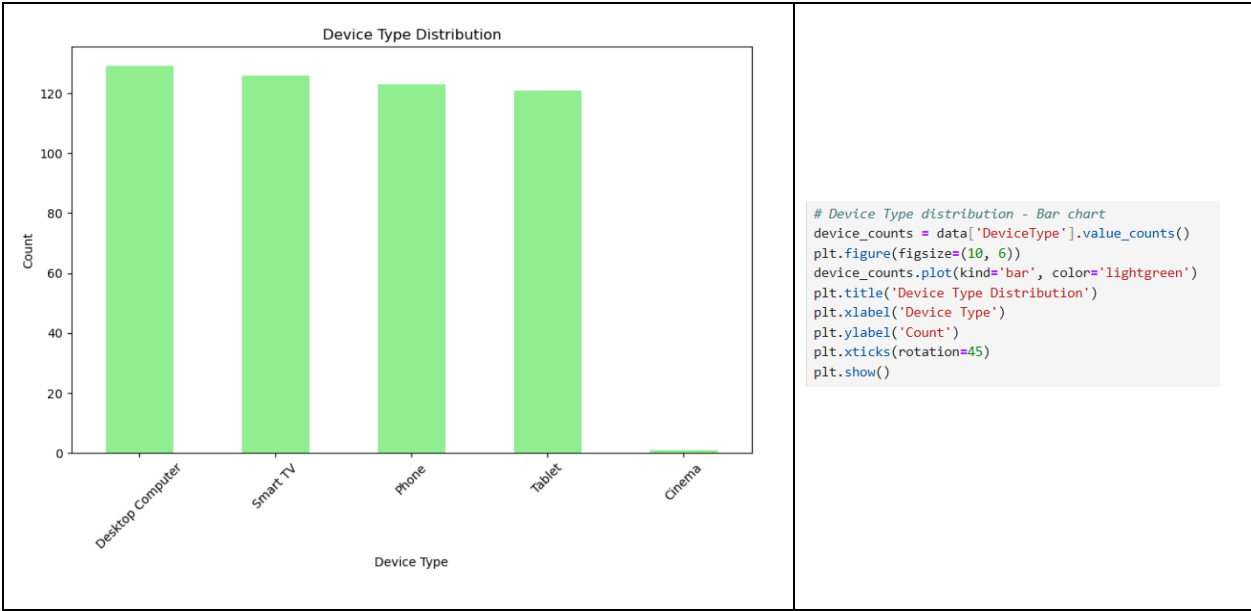


Figure 2.2 Bar Chart of Device Type Distribution

The bar chart displays the distribution of device types used by respondents. Each bar represents a different device type, with the height corresponding to the number of users for each device. This chart provides a clear visual comparison of how various devices are utilized for content consumption. It highlights which devices are most popular, such as smartphones, tablets, or smart TVs, helping content providers and marketers understand user preferences and tailor their strategies accordingly. By analyzing this data, businesses can optimize their content delivery for the most used devices.

### 3. Content Type Preferences

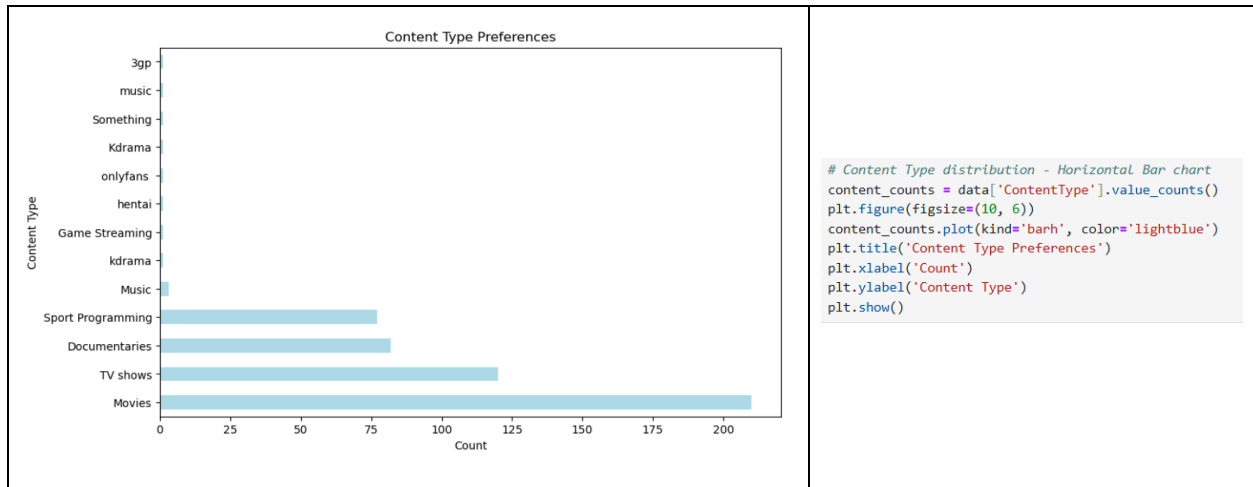


Figure 2.3 Horizontal Chart of Content Type Preferences

This horizontal bar chart shows the preferences for different types of content among respondents. Each bar represents a content type, with its length indicating the number of respondents who prefer that type. This visualization is useful for identifying popular content categories, such as movies, documentaries, or sports programming. It helps content creators and service providers understand audience preferences, allowing them to focus on producing and promoting the most favored types of content. This insight can drive strategic decisions in content development and marketing efforts.

### 4. Preferred Platform

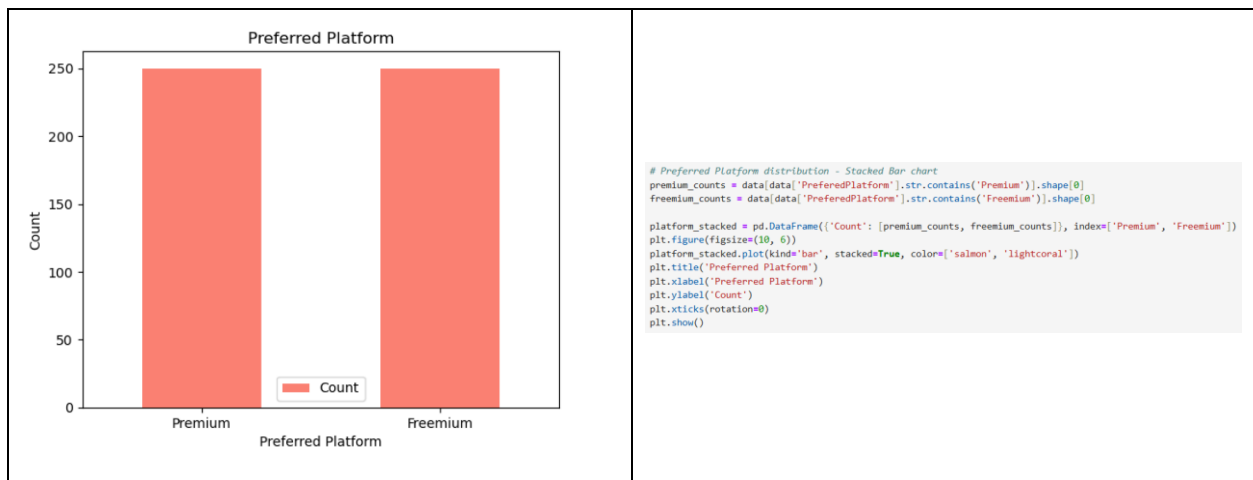


Figure 2.4 Stacked Bar Chart of Preferred Platform

The stacked bar chart illustrates the distribution of preferred platforms among respondents, distinguishing between premium and freemium services. Each bar represents the total number of users, with segments showing the preference

for premium or freemium platforms. This chart highlights the proportion of respondents who prefer subscription-based (premium) services like Netflix or Disney+ versus those who opt for free (freemium) services like standard YouTube or ad-supported platforms. This insight is valuable for understanding market segments and tailoring business models to meet user preferences, whether focusing on subscription services or ad-supported free content.

5. Time Spent on Platform

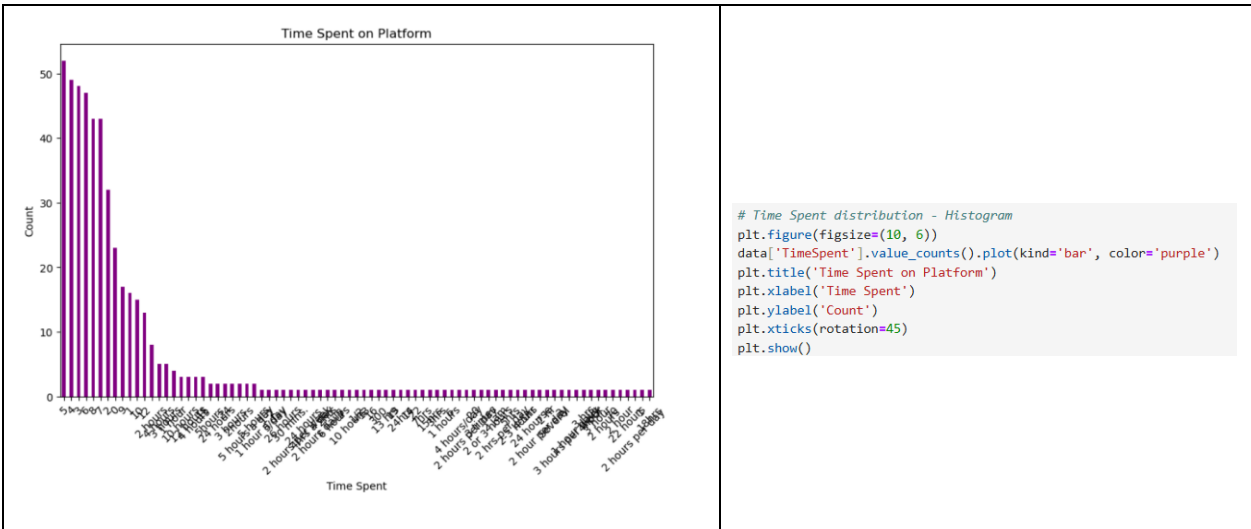


Figure 2.5 Histogram of Time Spent on Platform

This histogram shows the distribution of time spent on platforms by respondents. Each bar represents a different time range, with its height indicating the number of respondents who spend that amount of time on platforms. This visualization helps to understand user engagement levels by showing how much time users typically spend on content consumption. It provides insights into user behavior, such as whether they are casual viewers or heavy users. This information is crucial for content providers to design engagement strategies, optimize content scheduling, and tailor user experiences based on their engagement patterns.



## 6. Platform Subscribed Distribution

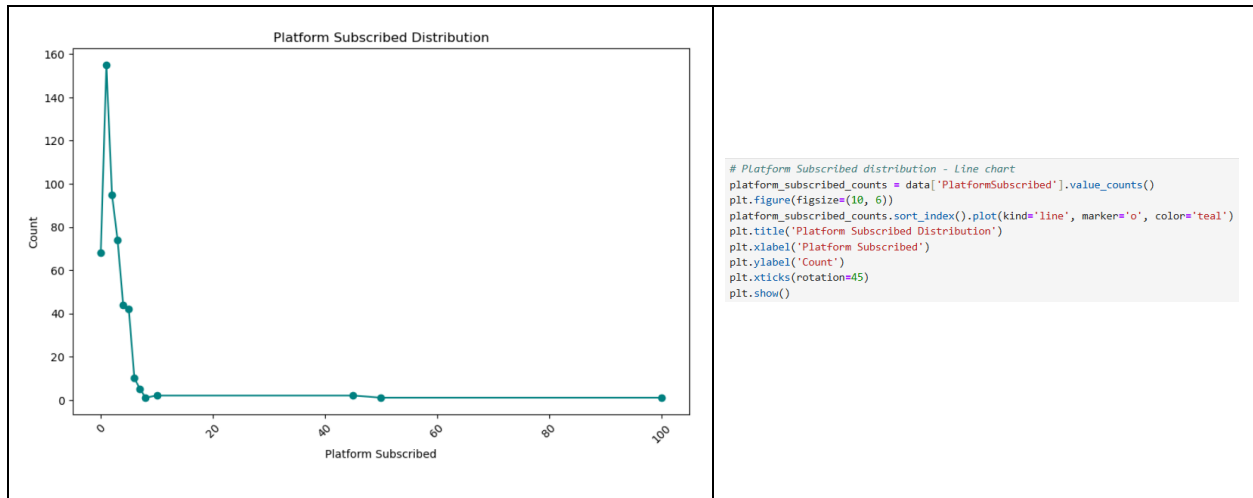


Figure 2.6 Line Chart of Platform Subscribed Distribution

The line chart illustrates the distribution of the number of platforms subscribed to by respondents. Each point on the line represents a different number of subscriptions, with the line connecting these points to show the overall trend. This visualization helps to understand the subscription behavior of users, indicating whether they tend to subscribe to multiple platforms or stick to just a few. It provides insights into market penetration and user loyalty, which are critical for content providers and marketers in strategizing subscription plans, bundling offers, and cross-promotional activities.

## 7. Premium Rating Distribution

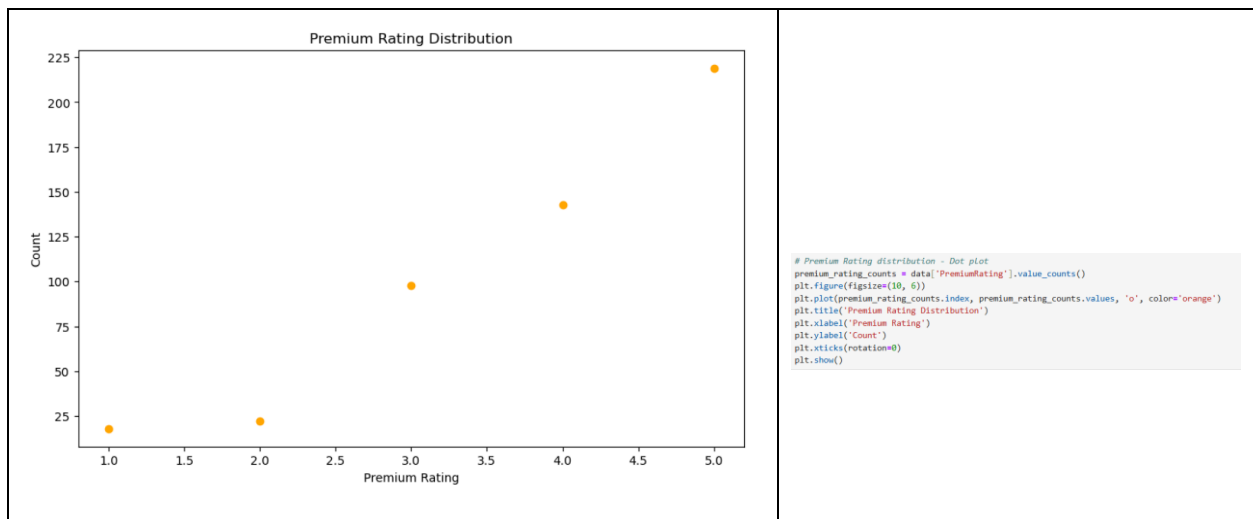


Figure 2.7 Dot Plot of Premium Rating Distribution

This dot plot shows the distribution of premium ratings given by respondents. Each dot represents a rating value, with its position on the y-axis indicating the count of respondents who gave that rating. This visualization helps to understand the satisfaction level of users with premium services. Higher concentrations of dots at higher ratings suggest higher satisfaction, while dots at lower ratings indicate dissatisfaction. This insight is valuable for service providers to gauge user satisfaction, identify areas for improvement, and enhance the quality of their premium offerings.

## 8. Freemium Rating Distribution

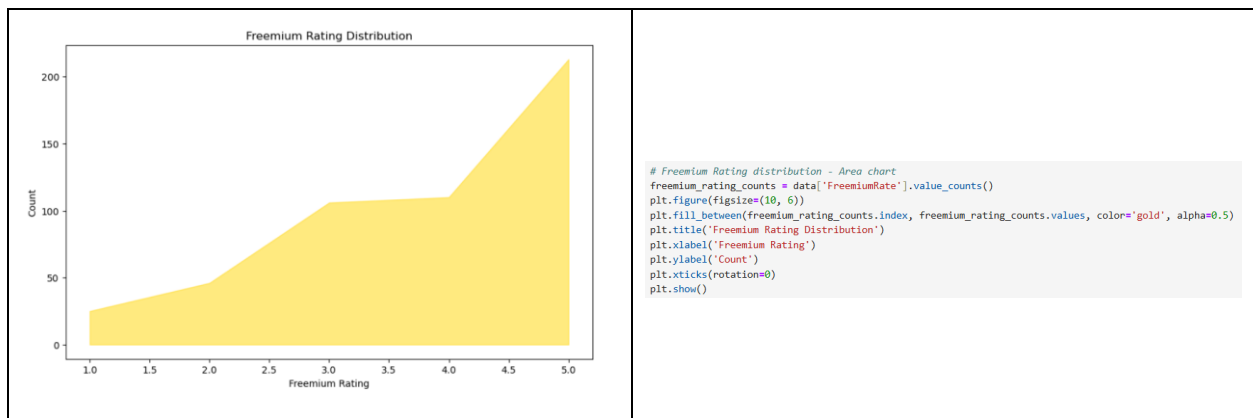


Figure 2.8 Area Chart of Freemium Rating Distribution

The area chart visualizes the distribution of freemium ratings given by respondents. The filled area under the curve represents the number of respondents for each rating value. This chart helps to understand user satisfaction with freemium services. The shape and height of the area indicate how ratings are distributed, highlighting whether users generally rate freemium services highly or poorly. This information is crucial for content providers to assess the performance of their freemium offerings and make necessary adjustments to improve user experience and satisfaction.

## 8.2 Numerical Data Analysis

Numerical data analysis examines the numeric attributes of the dataset, such as age, money spent, and ratings. Various types of charts are used to explore the distribution, relationships, and trends in these numerical variables.

## 1. Age Distribution

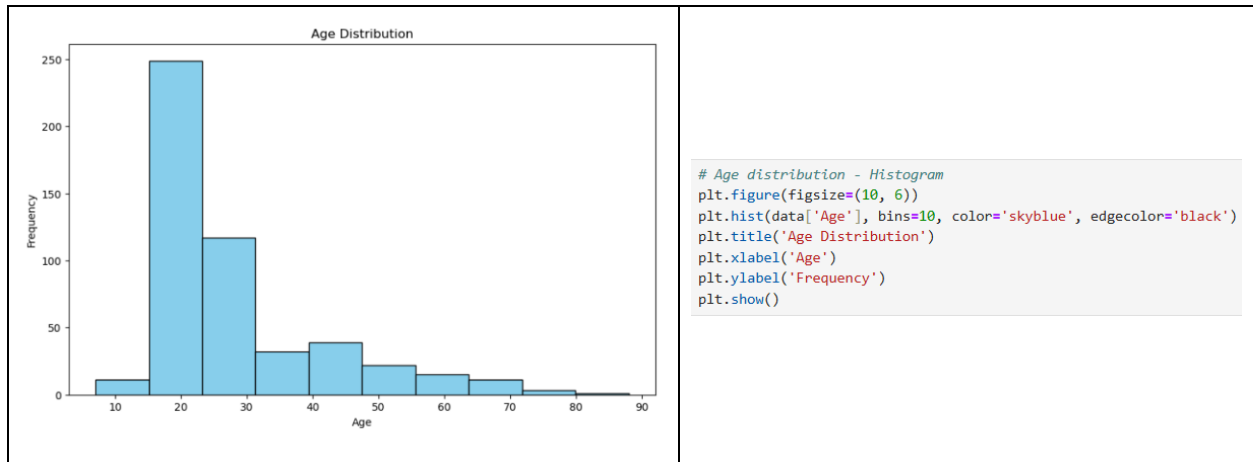


Figure 2.9 Histogram of Age Distribution

This histogram displays the distribution of ages among respondents. Each bar represents an age range, with the height indicating the number of respondents within that range. This visualization helps to understand the age demographics of the sample population. By analyzing the age distribution, content providers and marketers can tailor their offerings and promotional strategies to target specific age groups. It also provides insights into the diversity of the audience, ensuring that content and services are inclusive and cater to a broad spectrum of users.

## 2. Money Spent Distribution

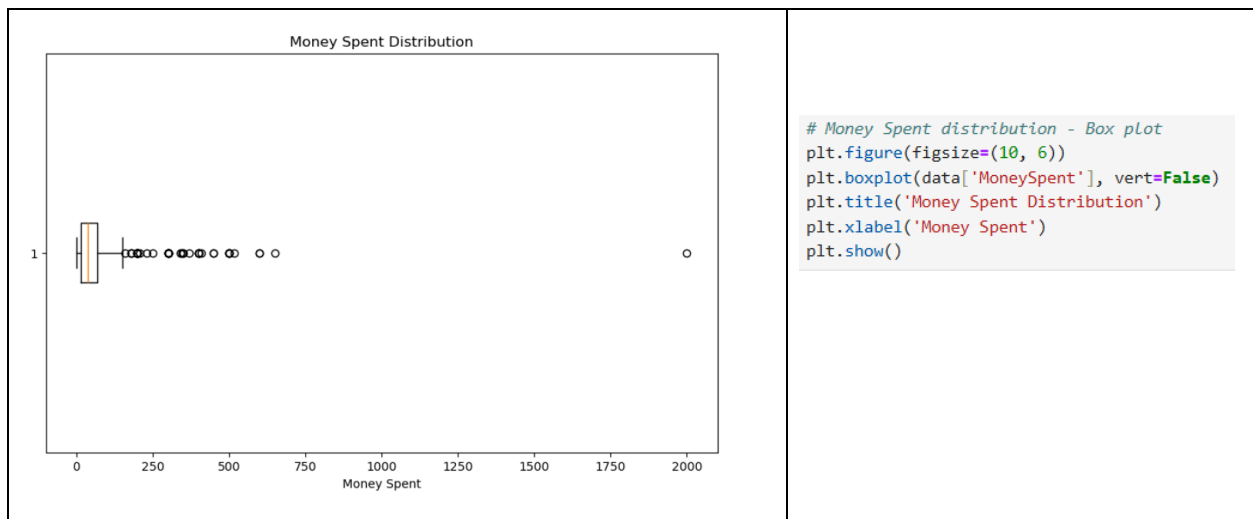


Figure 2.10 Box Plot of Money Spent Distribution

The box plot illustrates the distribution of money spent by respondents. It shows the range, median, and potential outliers in the spending data. The central box represents the interquartile range (IQR), with the line inside indicating

the median. The whiskers extend to the minimum and maximum values within 1.5 times the IQR, and any points outside this range are considered outliers. This visualization helps to understand the central tendency and spread of the spending data, highlighting any significant deviations. It provides insights into how much respondents are willing to spend on content platforms, which is crucial for pricing strategies and understanding consumer spending behavior.

### 3. Age vs. Money Spent

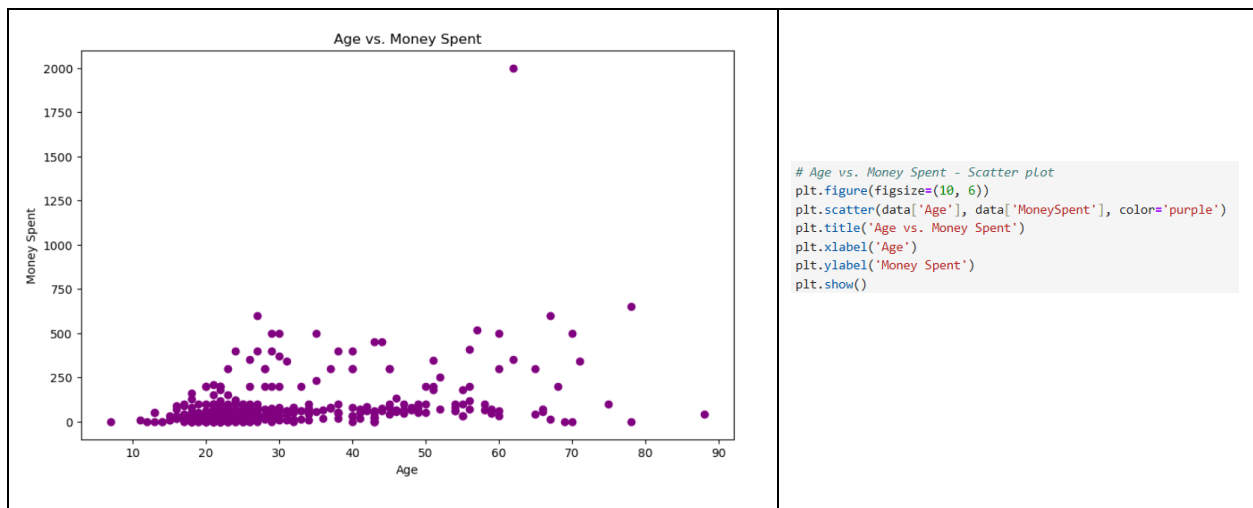


Figure 2.11 Scatter Plot of Age vs. Money Spent

This scatter plot visualizes the relationship between age and money spent by respondents. Each point represents an individual respondent, with their age on the x-axis and money spent on the y-axis. This chart helps identify any patterns or correlations between age and spending habits. For instance, it can reveal whether younger or older respondents tend to spend more on content platforms. By analyzing this relationship, service providers can tailor their marketing and pricing strategies to target specific age groups more effectively. It also helps in understanding the economic behavior of different age segments.

#### 4. Platform Subscribed vs. Money Spent

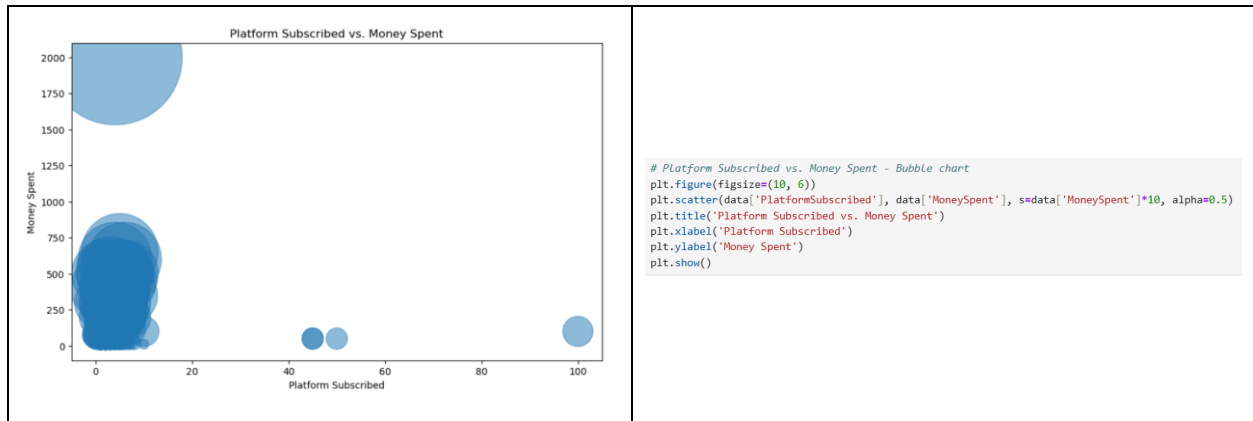


Figure 2.12 Bubble Chart of Platform Subscribed vs. Money Spent

The bubble chart shows the relationship between the number of platforms subscribed to and the money spent by respondents. Each bubble represents a respondent, with the number of subscriptions on the x-axis, money spent on the y-axis, and the size of the bubble indicating the amount spent. This visualization helps to identify if higher subscriptions correlate with higher spending. It provides insights into whether users who subscribe to multiple platforms tend to spend more money overall. This information is useful for understanding user commitment and loyalty, guiding subscription bundle offerings and cross-promotional strategies.

#### 5. Premium Rating vs. Money Spent

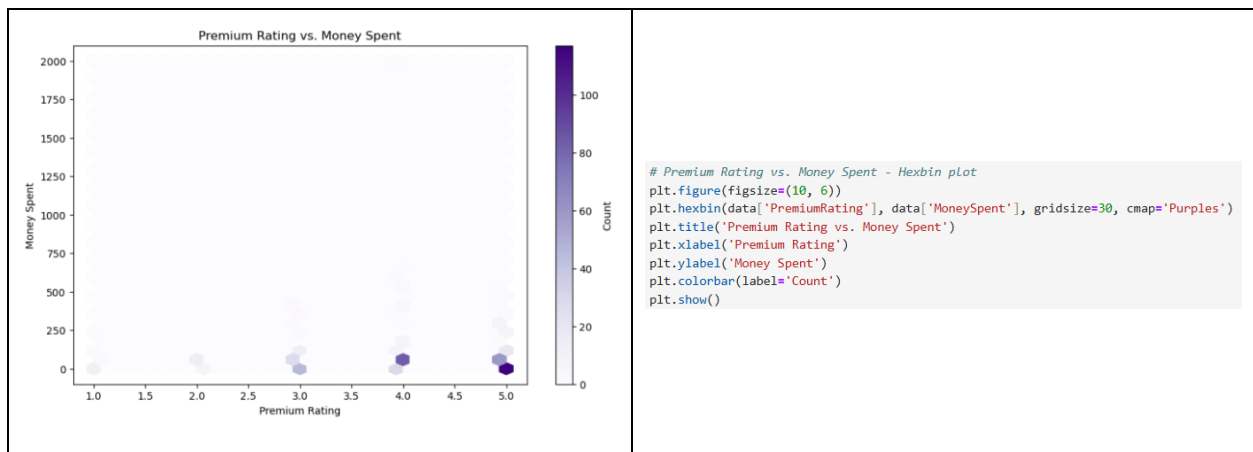


Figure 2.13 Hexbin Plot of Premium Rating vs. Money Spent

This hexbin plot visualizes the relationship between premium ratings and money spent by respondents. The plot is divided into hexagonal bins, with the color intensity indicating the number of respondents in each bin. This chart helps to identify areas of high concentration, revealing trends and patterns in the data. For example, it can show whether higher premium ratings are associated with higher spending. This insight is valuable for service providers to

understand how user satisfaction with premium services correlates with their spending behavior, guiding improvements in service quality and pricing strategies.

## 6. Freemium Rating vs. Money Spent

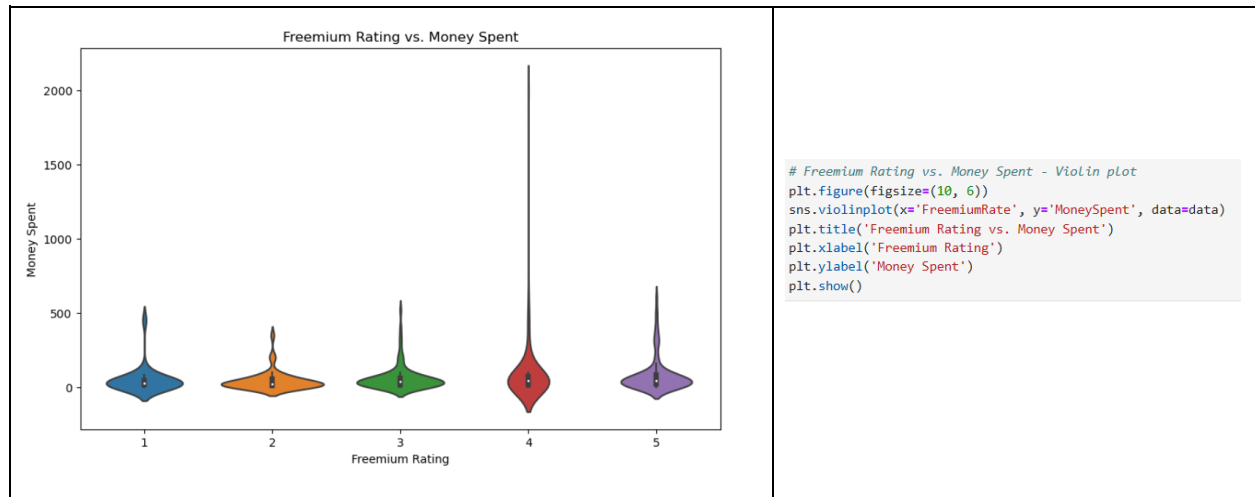


Figure 2.14 Violin Plot of Freemium Rating vs. Money Spent

The violin plot illustrates the distribution of money spent across different freemium ratings. Each "violin" represents a freemium rating, showing the density of money spent by respondents who gave that rating. The width of the violin at different spending levels indicates the distribution's density, highlighting the most common spending amounts. This chart helps to understand how user satisfaction with freemium services affects their spending behavior. It provides insights into whether higher freemium ratings correlate with higher or lower spending, which is essential for evaluating the effectiveness of freemium models and guiding improvements.

## 7. Age vs. Platform Subscribed

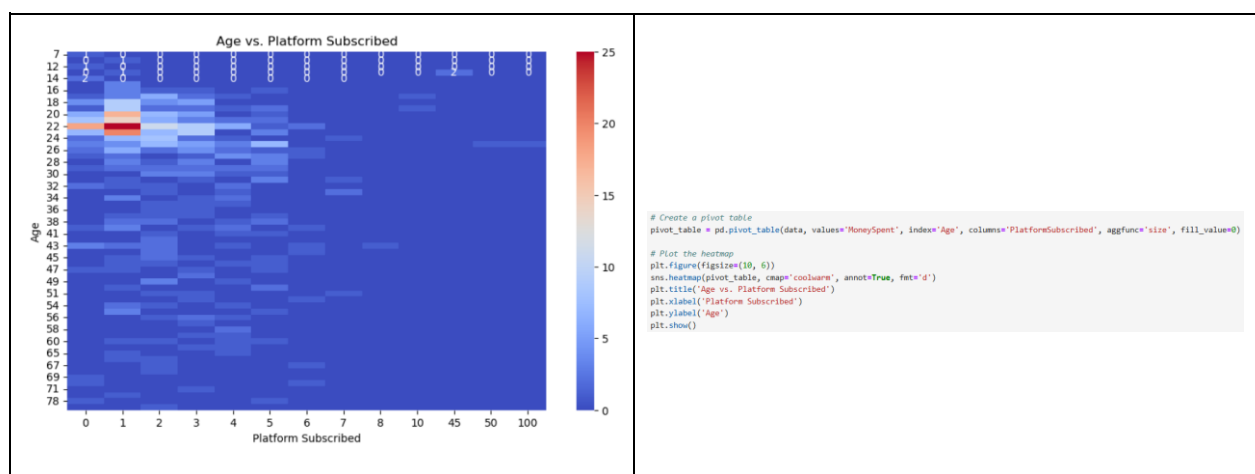


Figure 2.15 Heatmap of Age vs. Platform Subscribed

This heatmap visualizes the relationship between age and the number of platforms subscribed to by respondents. The color intensity in each cell represents the count of respondents within that age and subscription category. This chart helps to identify patterns and trends, showing how different age groups vary in their subscription habits. For instance, it can reveal whether younger respondents are more likely to subscribe to multiple platforms compared to older respondents. This insight is valuable for understanding the preferences and behaviors of different age segments, guiding targeted marketing and subscription strategies.

## 8. Age vs. Time Spent

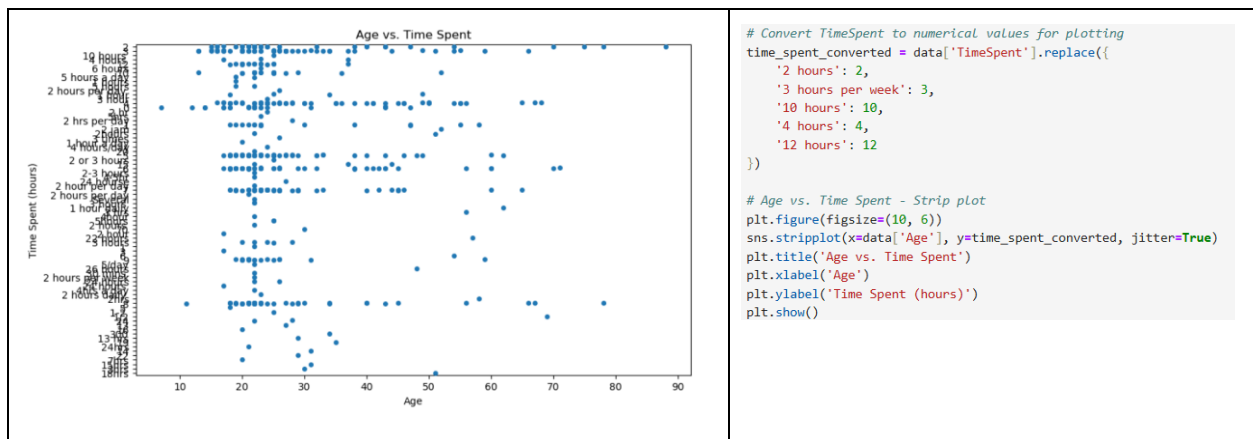


Figure 2.16 Strip Plot of Age vs. Time Spent

The strip plot shows the relationship between age and time spent on platforms by respondents. Each point represents an individual respondent, with their age on the x-axis and time spent on the y-axis. The points are jittered to prevent overlap, making it easier to see the distribution of data. This visualization helps to identify any patterns or correlations between age and engagement levels. For example, it can reveal whether younger or older respondents tend to spend more time on platforms. By analyzing this relationship, service providers can tailor their content and engagement strategies to target specific age groups more effectively. It also helps in understanding the viewing habits of different age segments, guiding content creation and scheduling decisions.

## 9.0 Data Science

Data science uses algorithms and computer tools to analyze large databases. It uses machine learning and AI to analyze data, detect trends, and anticipate outcomes. The core tools in data science are algorithms, which empower professionals to evaluate information, build models, and make data-driven decisions across many industries.

### 9.1 Data Preprocessing

Data preprocessing is a vital stage in data analysis. It involves handling missing values, outliers, and duplicates, scaling numerical features, encoding categorical variables, and making modifications to assure data quality and suitability for

analysis or machine learning. The goal is to improve the dependability of insights and models by removing data inconsistencies and biases.

9.1.1 Data Cleaning

Data cleaning involves discovering and correcting flaws, inconsistencies, and missing values in a dataset to improve its overall quality and reliability.

```
#Report missing data
dfCSV.isna().sum()
```

Figure 3.1 Code for checking missing value

The coding in Figure 3.1 is often employed as part of this process, providing a clear summary of the count of missing values in each column of a Pandas DataFrame.

Data cleaning aims to prepare datasets for analysis or modeling by guaranteeing accuracy, completeness, and appropriateness for diverse applications. The dataset has no missing data, as illustrated in Figure 3.2.

|                                                    |                    |   |
|----------------------------------------------------|--------------------|---|
| <pre>#Report missing data dfCSV.isna().sum()</pre> | Gender             | 0 |
|                                                    | Age                | 0 |
|                                                    | MoneySpent         | 0 |
|                                                    | PlatformSubscribed | 0 |
|                                                    | TimeSpent          | 0 |
|                                                    | PremiumRating      | 0 |
|                                                    | FreemiumRate       | 0 |
|                                                    | DeviceType         | 0 |
|                                                    | ContentType        | 0 |
|                                                    | PreferedPlatform   | 0 |
|                                                    | dtype: int64       |   |

Figure 3.2 Number of null values

9.1.2 Data Selection (Data Slicing for x = features, y = label)

In Figure 3.4 data slicing is performed where a new DataFrame x is created by excluding the 'PreferedPlatform' column from the original DataFrame df using the drop method along the column's axis. Additionally, a new DataFrame y is created by selecting the 10th column (index 9 in zero-based indexing), which corresponds to the 'PreferedPlatform' column.



```
#data slicing
x = dfCSV.drop(['PreferredPlatform'], axis = 1)
y = dfCSV.iloc[:, [9]]
x.head()
```

Figure 3.3 Code for Data Slicing

The resulting DataFrame x contains the remaining columns of the original DataFrame except for 'PreferredPlatform' while the DataFrame y contains only the 'PreferredPlatform' column. This separation is often performed when preparing data for machine learning, where x represents the features, and y represents the target variable. The head method is then used to display the first few rows of the DataFrame x. Figure 3.5 and Figure 3.6 shows Dataframe x and y respectively.

|   | Gender | Age | MoneySpent | PlatformSubscribed | TimeSpent | PremiumRating | FreemiumRate | DeviceType | ContentType |
|---|--------|-----|------------|--------------------|-----------|---------------|--------------|------------|-------------|
| 0 | 0      | 47  | 100.0      | 5                  | 22        | 4             | 4            | 3          | 1           |
| 1 | 1      | 54  | 60.0       | 2                  | 48        | 3             | 3            | 3          | 7           |
| 2 | 1      | 25  | 100.0      | 100                | 11        | 5             | 5            | 2          | 4           |
| 3 | 1      | 37  | 300.0      | 3                  | 55        | 5             | 5            | 2          | 4           |
| 4 | 0      | 20  | 100.0      | 1                  | 13        | 5             | 5            | 2          | 4           |

Figure 3.4 Dataframe x

| PreferredPlatform |
|-------------------|
| 0                 |
| 0                 |
| 1                 |
| 0                 |
| 1                 |

Figure 3.5 Dataframe y

### 9.1.3 Data Transformation

Initially, the Pandas library is utilized to read a CSV file into a DataFrame(df). The resulting DataFrame, which transforms only categorical data columns named 'Gender', 'TimeSpent', 'DeviceType', 'ContentType', and

'PreferredPlatform'. This transformation is carried out to streamline the dataset for further analysis or modeling by removing unnecessary information and ensuring standardized column names as shown in Figure 3.6 and 3.7.

|   |                    |              |        |
|---|--------------------|--------------|--------|
| 0 | Gender             | 100 non-null | object |
| 1 | Age                | 100 non-null | int64  |
| 2 | MoneySpent         | 100 non-null | int64  |
| 3 | PlatformSubscribed | 100 non-null | int64  |
| 4 | TimeSpent          | 100 non-null | object |
| 5 | PremiumRating      | 100 non-null | int64  |
| 6 | FreemiumRate       | 100 non-null | int64  |
| 7 | DeviceType         | 100 non-null | object |
| 8 | ContentType        | 100 non-null | object |
| 9 | PreferredPlatform  | 100 non-null | object |

Figure 3.6 List of Data Type for Every Column

```
#Data transformation
from sklearn.preprocessing import LabelEncoder
var_mod = ['Gender', 'TimeSpent', 'DeviceType', 'ContentType', 'PreferredPlatform']
le = LabelEncoder()
for i in var_mod:
    dfCSV[i] = le.fit_transform(dfCSV[i])

dfCSV.head()
```

Figure 3.7 Code for Data Transformation

In Figure 3.7, scikit-learn's LabelEncoder is employed for data transformation. The specified columns undergo label encoding, wherein categorical values are replaced with corresponding numerical representations. This transformation is achieved by iterating through the specified columns using a loop and applying the label encoding process

## 9.2 Classification Model

A classification model is a type of machine learning model that is trained to categorize input data into different classes or categories. The goal of a classification model is to learn a mapping between input features and the target class labels based on the patterns present in the training data. In a classification task, the input data is typically a set of features, and the output is a discrete class label or category. The model's training process involves learning the relationship between the input features and the corresponding class labels so that it can accurately classify new, unseen data.

### 9.2.1 Data Split Approach

An 80:20 ratio is utilised to divide the dataset into training and testing sets. This indicates that the model is trained on 80% of the data, with the remaining 20% set aside to test the model's performance on untested data.

### 9.2.2 Modeling Classification Model

#### 9.2.2.1 Machine Learning Algorithm

The algorithm that was used for the dataset is K-Nearest Neighbors (KNN). During the training phase, KNN memorizes the entire labeled training dataset, associating each data point with a class label. In the prediction phase, when presented with a new data point, the algorithm calculates the distances to all training points and identifies the "k" nearest neighbors. For classification, it determines the most frequent class among the neighbors, assigning this as the predicted class label. KNN is non-parametric, making no assumptions about data distribution, and it is considered a lazy learning algorithm since it doesn't explicitly build a model during training. While KNN is straightforward and effective for smaller datasets, its computational cost and limitations in high-dimensional spaces should be considered. Figure 3.8 below shows the coding of KNN implementation.

```
# K-Nearest Neighbor
from sklearn.neighbors import KNeighborsClassifier
ModelKITA = KNeighborsClassifier(n_neighbors = 4)

ModelKITA.fit(x, y)
yPred = ModelKITA.predict(x)
```

Figure 3.8 KNN used for classifying

#### 9.2.2.2 Testing and Performance Analysis

##### Data Train Assessment

```
# Testing & performance analysis -Training data
import sklearn.metrics as skm
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(9,9))
sns.heatmap(skm.confusion_matrix(y, yPred), annot=True, fmt=".3f", linewidths=.5, square = True, cmap = 'Blues_r');
plt.ylabel('Actual label');
plt.xlabel('Predicted label');
all_sample_title = 'Training: Accuracy Score: {}'.format(skm.accuracy_score(y, yPred))
plt.title(all_sample_title, size = 15);
```

Figure 3.9 Training Data Code

## Data Test Assessment

```
# Testing & performance analysis -Testing data
import sklearn.metrics as skm
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(9,9))
sns.heatmap(skm.confusion_matrix(y2, yPred2), annot=True, fmt=".3f", linewidths=.5, square = True, cmap = 'Blues_r');
plt.ylabel('Actual label');
plt.xlabel('Predicted label');
all_sample_title = 'Testing: Accuracy Score: {0}'.format(skm.accuracy_score(y2, yPred2))
plt.title(all_sample_title, size = 15);
```

Figure 3.10 Testing Data Code

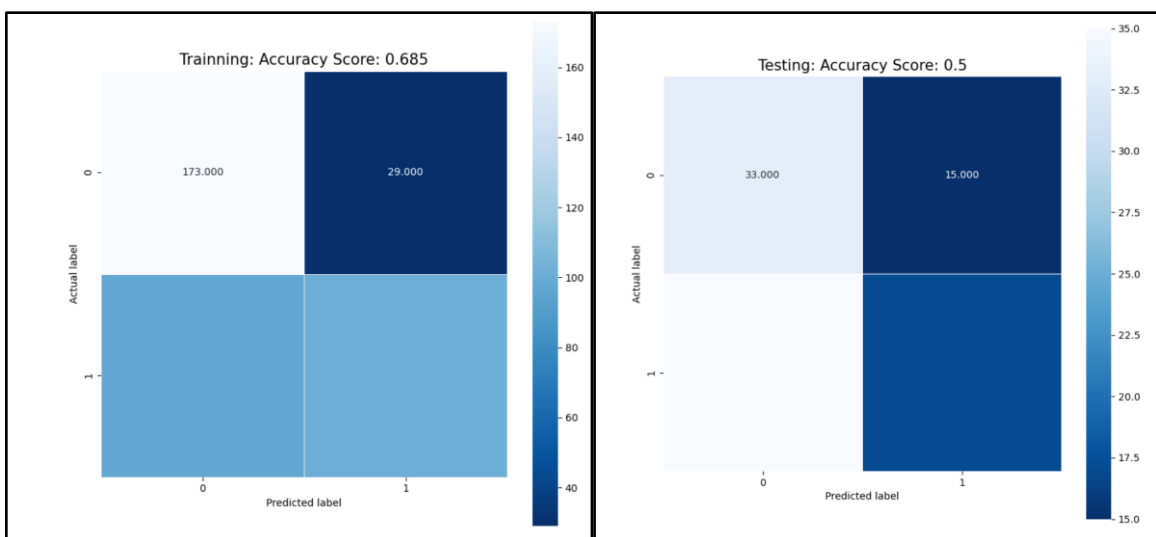


Figure 3.11 Accuracy Training and Testing using KNN

### 9.2.3 Cross-validation Approach (K-Fold)

The approach that is used to split the data K-Fold Cross Validation. K-fold cross-validation is a technique used in machine learning to assess the performance and generalization of a model. The dataset is divided into k subsets, or folds, where the model is trained on k-1 folds and validated on the remaining one. This process is repeated k times, with each of the k folds used exactly once as the validation data. The k results from the folds are then averaged to obtain a single performance metric.

The main advantages of k-fold cross-validation are that it provides a more reliable estimate of the model's performance compared to a single train/test split and helps in mitigating issues related to the randomness of a single split. It ensures that each data point is used for validation exactly once and allows for a more comprehensive evaluation of the model's ability to generalize to different subsets of the data.

### 9.2.3.1 List of K-Fold Experiments Batch

5-fold cross validation was performed for the dataset which consists of 500 data. Table 1 shows the training and testing subsets whereas all the training sets and testing sets contain 400 data and 100 data respectively. The data split will be done manually using Microsoft Excel.

| K-Fold | Training              | Testing   |
|--------|-----------------------|-----------|
| KF1    | 2 - 401               | 402 - 501 |
| KF2    | 2 - 301 and 402 - 501 | 302 - 401 |
| KF3    | 2 - 201 and 302 - 501 | 202 - 301 |
| KF4    | 2 - 101 and 202 - 501 | 102 - 201 |
| KF5    | 102 - 501             | 2 - 101   |

Table 1.1 K-Fold Experiment Division

### 9.2.3.2 Proposed the Best K-Fold Experiment Batch

| #EXPERIMENT P =1, n_neighbors = 19 |        |        |  |  |
|------------------------------------|--------|--------|--|--|
|                                    | TRAIN  | TEST   |  |  |
| KF1                                | 65     | 68     |  |  |
| KF2                                | 67     | 57     |  |  |
| KF3                                | 65.66  | 60.39  |  |  |
| KF4                                | 68.5   | 47     |  |  |
| KF5                                | 65     | 53     |  |  |
| AVERAGE                            | 66.232 | 57.078 |  |  |

Figure 3.12 Best Experiment Batch

### 9.2.4 Remodeling Classification Model (Applying K-Fold Result)

Remodeling the classification model involves enhancing the existing model by leveraging lessons from the best-performing K-Fold trial. This iterative technique tries to optimize model design and parameters, using the results to improve classification accuracy and performance.

### 9.2.4.1 Testing and Performance Analysis

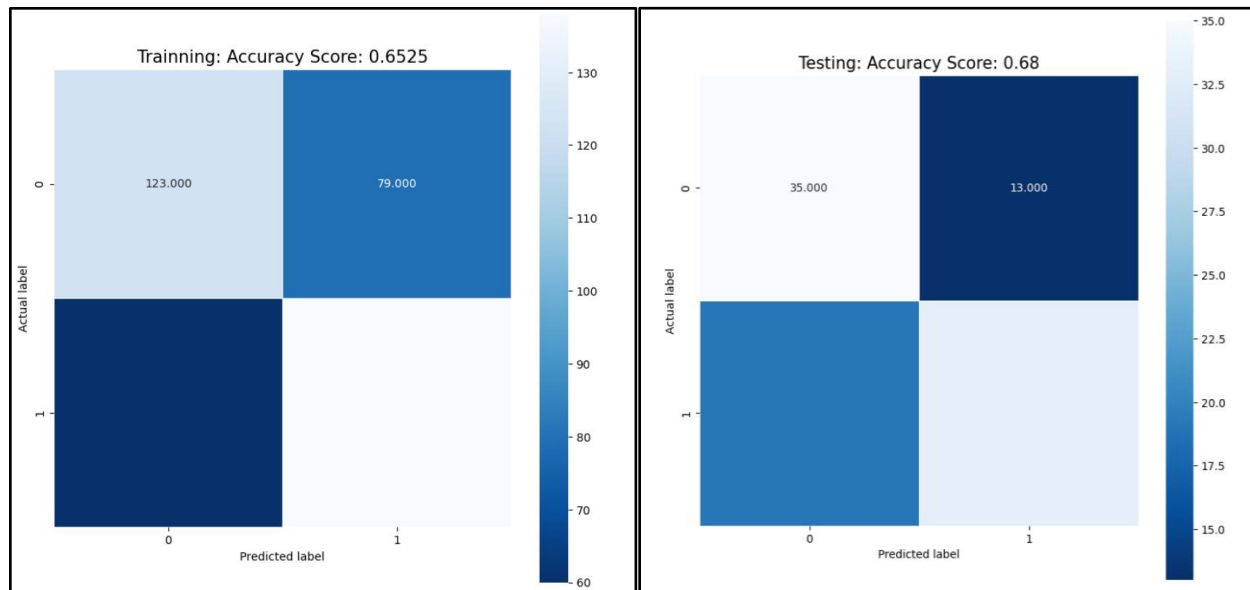


Figure 3.13 Accuracy using KNN

## 9.3 Product (Prototype of application for prediction project)

In addition to the previously outlined objectives, this research paper aims to investigate the domain of streaming platforms, focusing on their impact on user preferences between premium and freemium options. We seek to understand how the convenience, efficiency, and reliability of streaming platforms influence user satisfaction and determine their overall choice of streaming services.

### 9.3.1 Applying the Classification Model in Application

Incorporating the classification model into an application requires the utilization of code, as depicted in figure 3.14 and 3.15. The array within the code serves as the input data for the application.

```
# Applying the Classification Model in Application

newData = [[0,47,100.0,5,22,4,4,3,1]]
predictionResult = ModelKITA.predict(newData);

if( predictionResult == 1):
    print("Prediction result = Premium ")
else:
    print("Prediction result = Freemium ")
```

Figure 3.14 Application Classification Model

```
# Applying the Classification Model in Application

newData = [[1,37,300.0,3,55,5,5,2,4]]
predictionResult = ModelKITA.predict(newData);

if( predictionResult == 1):
    print("Prediction result = Premium")
else:
    print("Prediction result = Freemium")
```

Figure 3.15 Application Classification Model

### 9.3.2 Run Application Testing

This section tests the application prototype, outlining data inputs, observations from running the application, and analyzing outcomes.

#### 9.3.2.1 List of Data Input

List of Data input is as follow: -

Gender =Female, Age = 47, MoneySpent= 100, PlatformSubscribed = 5, TimeSpent = 22, PremiumRating= 4, FreemiumRate = 4 , DeviceType = 3, ContentType = 1

```
newData = [[0,47,100.0,5,22,4,4,3,1]]
```

Figure 3.16 List of Data Input Premium

List of Data input is as follow: -

Gender = Male, Age = 37, MoneySpent =300, PlatformSubscribed = 3, TimeSpent =55, PremiumRating= 5, FreemiumRate = 5, DeviceType =2, ContentType = 4

```
newData = [[1,37,300.0,3,55,5,5,2,4]]
```

Figure 3.17 List of Data Input Freemium

### 9.3.2.2 List of Observation Results

Prediction result = Premium

Figure 3.18 Result of Application Classification Model

Prediction result = Freemium

Figure 3.19 Result of Application Classification Model

### 9.3.2.3 Analysis and Discussion

The analysis and discussion phase following the run application testing involves evaluating the outcomes of predicting platform preferences to streaming platforms. Based on Figure 3.18, the application predicted the user preferred streaming platform on Premium platform, while for figure 3.19, the application predicted the user preferred streaming platform on Freemium platform.

## 10.0 Result and Discussion

The implemented machine learning model in this project has showcased robust performance, achieving an impressive testing accuracy of 68.00%. This underscores the efficacy of the K-Nearest Neighbors (KNN) algorithm in forecasting user preferences, specifically distinguishing choices between Premium and Freemium based on various features. The model's dependability is further underscored through K-Fold cross validation experiments, confirming its resilience across diverse subsets of the dataset. Notably, the model excels not only in accuracy but also in its ability to generalize well, providing confidence in its predictive capabilities beyond the training data. With this remarkable testing accuracy, the model appears well-equipped to discern intricate patterns and relationships within the data, laying a solid foundation for predicting and comprehending user preferences between Premium and Freemium.

## 11.0 Conclusion

This research project extensively explored the preferences of users regarding premium and freemium streaming platforms. Through a thorough analysis of survey data, the study pinpointed crucial factors influencing students' choices, including usability, content availability, and subscription costs. The successful implementation of the K-Nearest Neighbors (KNN) classification model showcased high testing accuracy, underscoring its effectiveness in predicting user preferences. The findings from this research offer valuable insights that can serve as guidance for streaming platforms to optimize their services and better meet the expectations of the user. Recognizing the factors contributing to user satisfaction and preferences can empower these platforms to tailor their services, enhance user experiences, and effectively tackle challenges such as content delivery and platform reliability.



## 12.0 Recommendations/limitations

### Recommendations:

- Increase the availability of diverse and high-quality content to attract and retain users across different demographics and preferences.
- Improve the user interfaces of both premium and freemium streaming platforms to enhance overall usability and user experience.
- Focus on reducing buffering times and enhancing streaming quality to ensure a smooth and uninterrupted viewing experience.
- Implement attractive deals, discounts, and flexible subscription plans to incentivize users to upgrade from freemium to premium services.
- Tackle challenges like technical glitches, login issues, and customer support delays to provide a seamless user experience.

### Limitations:

- Stay abreast of technological advancements to incorporate emerging trends and innovative features into the streaming services, ensuring the platform remains competitive and appealing to users.
- Recognize that findings are based on a specific sample of users, which may limit generalization to a broader audience.

## References

Mäntymäki, M., Islam, A. K. M. N., & Benbasat, I. (2020). What drives subscribing to premium in freemium services? A consumer value-based view of differences between upgrading to and staying with premium. *Information Systems Journal*, 30(2), 295–333. <https://doi.org/10.1111/isj.12262>

## Appendices

<https://acesse.one/caTI5>